

Elizabeth Pierburg

Atlanta, Georgia 30308 • elliepierburg@gmail.com • 678-725-6029

EDUCATION

Georgia Institute of Technology

Bachelor of Mechanical Engineering Major

Minor in Computing & Intelligence • Cumulative GPA: 3.96/4.0

Atlanta, Georgia

Anticipated Graduation May 2027

WORK EXPERIENCE

Yamaha Motor Manufacturing Corporation

Newnan, Georgia

Robotics/Mechatronics Co-Op

May 2025 to Present

- Developed embedded C++ control logic for an automatic brake system, integrating throttle position and motor-speed sensors to actuate brake calipers on Yamaha Golf Cars.
- Created 100+ Python automated validation tests for infotainment systems across three software platforms, reducing verification effort from approximately 120 to 40 labor-hours.
- Led AI-assisted CAD automation with Cursor, NX, and Teamcenter to generate 3,000+ standardized component models in one week, expanding Yamaha's reusable CAD library.
- Designed, fabricated, and validated Arduino-based test harnesses incorporating motor drivers, sensors, soldered electronics, and vehicle hardware for safety-critical system verification.
- Conducted bench- and vehicle-level testing using CAN communication, GPS validation tools, and automated robotic testing platforms, debugging issues through production development.
- Led environmental reliability testing of vehicle main switches; designed a 16-channel test harness for simultaneous salt-spray testing and analyzed electrical resistance data to investigate corrosion-related failures.

SAFE Robotics Lab

Atlanta, Georgia

Mobile Manipulator Platform Team

January 2025 to Present

- Integrated mechanical, electrical, and software subsystems for a mobile manipulator platform utilizing NVIDIA Jetson hardware and ROS.
- Developed and validated ROS-based motion-planning algorithms for coordinated robotic arm and mobile-base operation.
- Designed and prototyped a 3D-printed rocker pedal for foot-based robot control while supporting system integration with NVIDIA Jetson hardware and ROS-based robotics environments.

Robo Nav, Robojackets

Atlanta, Georgia

Project Manager, Mini Rover Challenge | Mechanical Science Member

August 2024 to Present

- Managed a six-month rover development project for new members, coordinating training, project scheduling, and technical milestones.
- Designed and manufactured rover components using SolidWorks, additive manufacturing, waterjet cutting, and laser cutting.
- Collaborated with multidisciplinary teams to integrate mechanical and science subsystems for autonomous mission execution.

Georgia Institute of Technology, Tutoring & Academic Support

Atlanta, Georgia

Learning Assistant/Grader, Calculus & Thermodynamics

May 2024 to December 2025

- Integrated effective, evidence-based tutoring techniques and strategies to student learning styles and needs.
- Assessed assignments and supported course delivery for 100+ students.

TECHNICAL SKILLS

Programming: Python, C++, C, MATLAB, Java, Git

Robotics & Controls: ROS, Arduino, Sensor Integration, CAN Communications, Embedded Systems

CAD/Engineering: NX, SolidWorks, Fusion, Simulink, Teamcenter

Manufacturing: 3D Printing, Water Jetting, Laser Cutting, Mechanical Assembly, Electrical Harness Design

Testing & Validation: Automated Testing, Hardware Validation, Reliability Testing, Root Cause Analysis

AI Productivity Tools: Cursor, GitHub Copilot

CERTIFICATES AND HONORS

- Scrum Master Certified (SSGI), 2026
- Zell Miller Scholar
- Dean's List, 2023 to Present
- Creative Designs and Decisions, Design Award Winner, Fall 2024